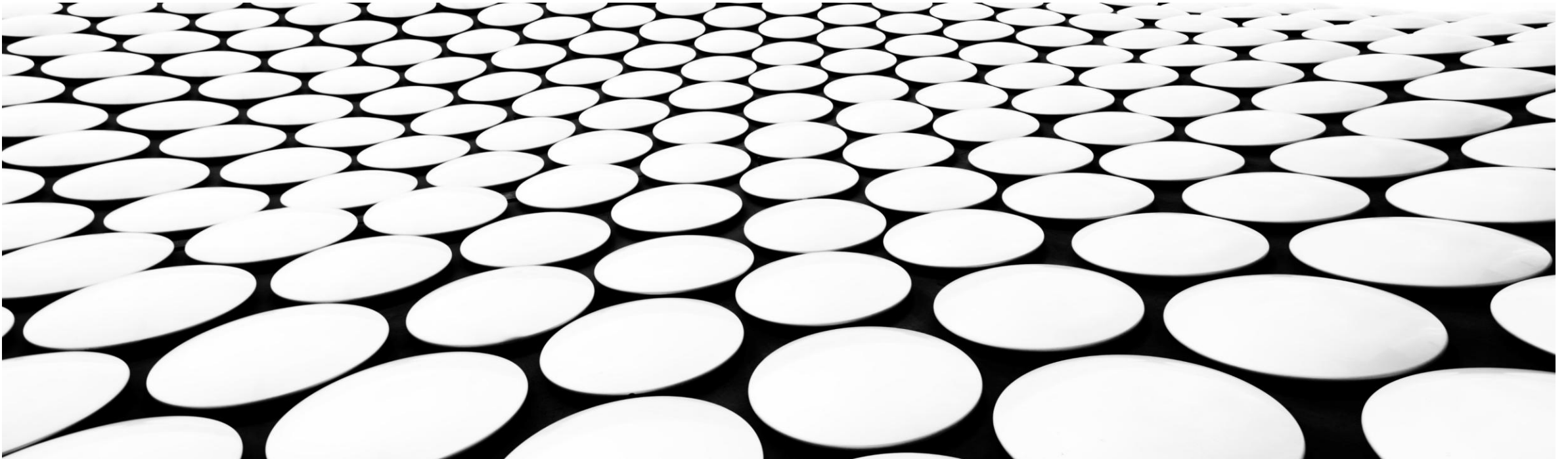
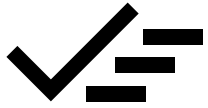


ENSEMBLE MODELS

Combining multiple predictions





ENSEMBLE MODELS

- Introduction
- Ensemble approaches:
 - Voting
 - Bagging
 - Stacking and Blending
 - Boosting

INTRODUCTION

What Is Ensemble Learning?



Article by Yulia Gavrilova
May 27th, 2024

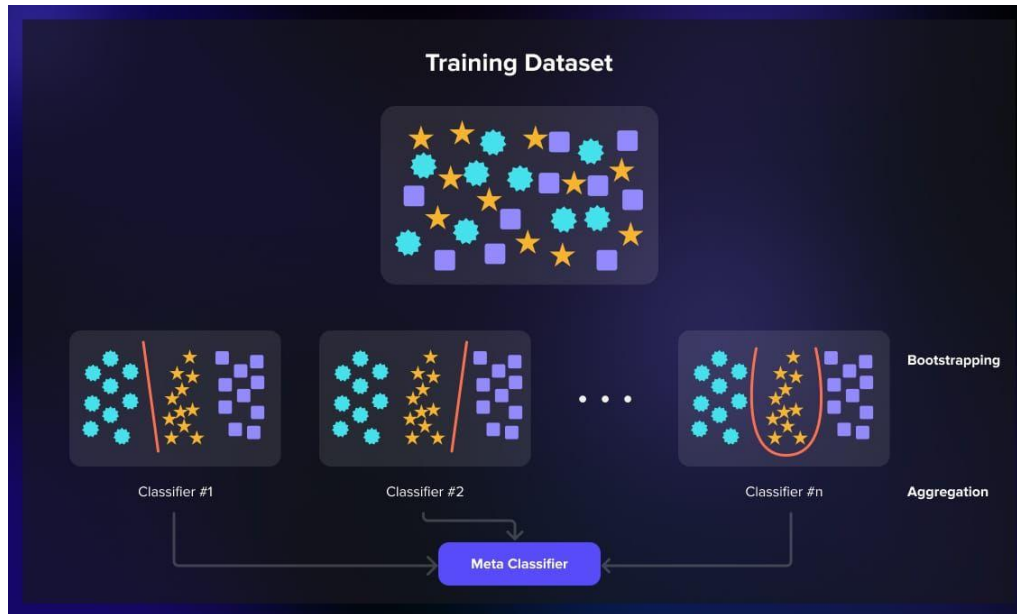
9 min read

[Source](#)

Ensemble learning is a machine learning paradigm that proposes to use multiple models to create a stronger model.

The fundamental idea behind ensemble learning is that a group of models, when working together, can outperform any individual model.

It is a lot like teamwork in a corporation: a team of specialists can potentially come up with a more qualified solution than an individual employee.



For ensemble learning to work, **you need to use diverse models**. If all models in the ensemble are similar, they will probably make similar errors.

Different models capture different aspects of the underlying data patterns, and their errors do not coincide. This way, when combined, the strengths of one model can compensate for the weaknesses of another.

Why Use Ensemble Techniques?

- **Reduce Overfitting:** By combining multiple models, ensemble techniques can smooth out irregularities that may cause overfitting in a single model.
- **Improve Accuracy:** Combining multiple weak models leads to a more accurate prediction.
- **Increase Stability:** Ensembles are more robust to variations in the training data and less sensitive to noise.

Decision trees can overfit

A Comprehensive Guide to Ensemble Techniques: Bagging and Boosting



Abhishek Jain · Follow
6 min read · Sep 13, 2024

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Ensemble Learning — Voting and Bagging with Python



Changhyun Kim · Follow
5 min read · Jul 15, 2022

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Different types of Ensemble Techniques — Bagging, Boosting, Stacking, Voting, Blending



Abhishek Jain · Follow
7 min read · Sep 13, 2024

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Ensemble Learning: Stacking, Blending & Voting

If you want to increase the effectiveness of your ML model, maybe you should consider Ensemble Learning

Fernando López
Dec 13, 2020 · 10 min read



[Source](#)

 SPOT INTELLIGENCE

Top 6 Most Powerful Ensemble Learning Techniques Explained & Algorithms That Implement Them

by Neri Van Otten | Aug 9, 2023 | Data Science, Machine Learning

[Source](#)



1. Voting
2. Bagging
3. Stacking and Blending
4. Boosting

1. Voting
2. Bagging
3. Stacking and Blending
4. Boosting

Voting

Definition

In **voting**, multiple models are trained independently on the same dataset, and their predictions are combined by **voting** in the case of classification tasks, or by **averaging** in the case of regression tasks. This is one of the simplest ensemble methods and can be classified into two types: **hard voting** and **soft voting**.

Hard Voting:

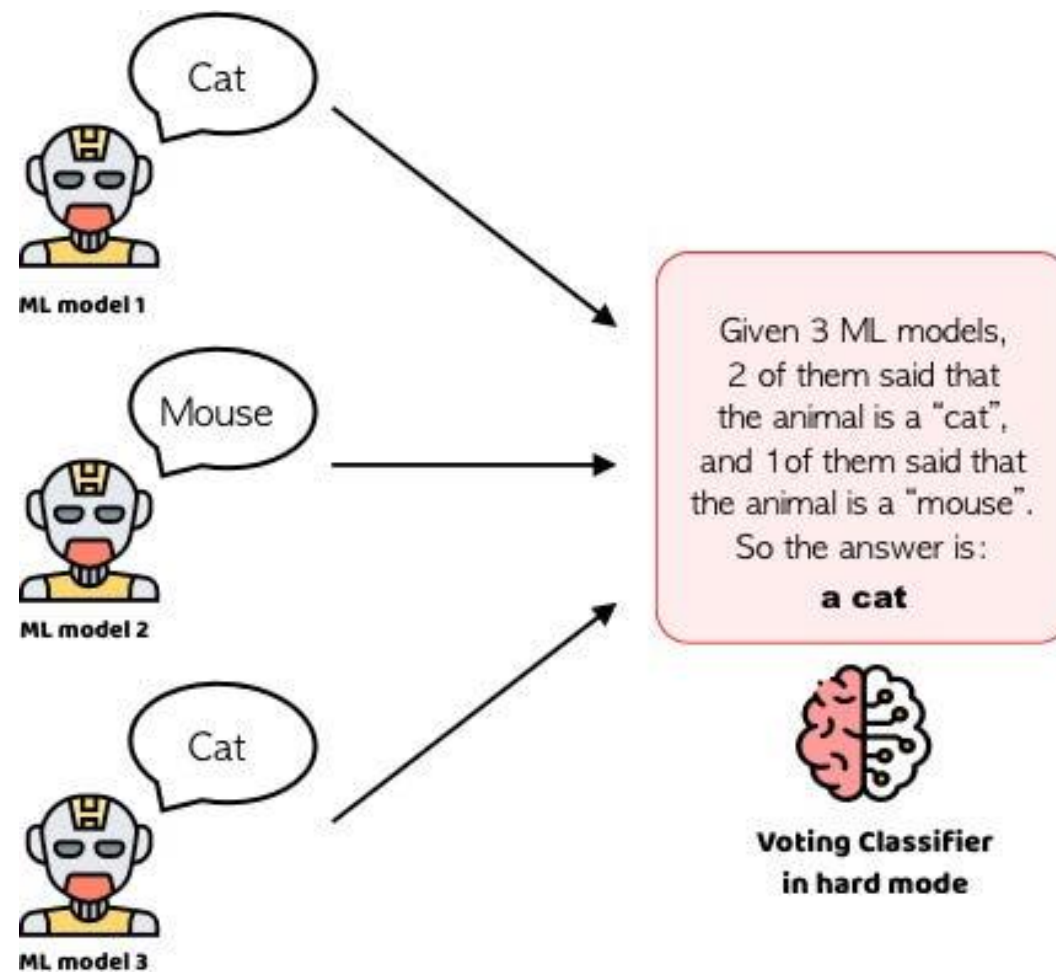
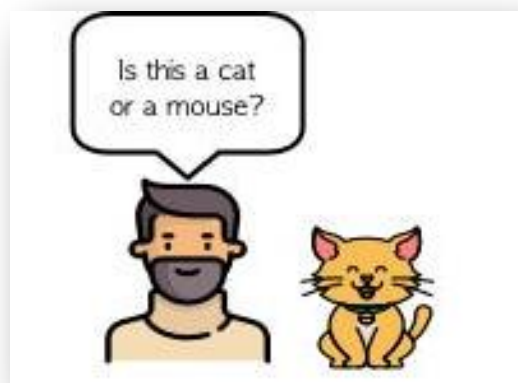
In classification tasks, the final ensemble prediction is determined by selecting the class that receives the most votes from the base models' predictions.

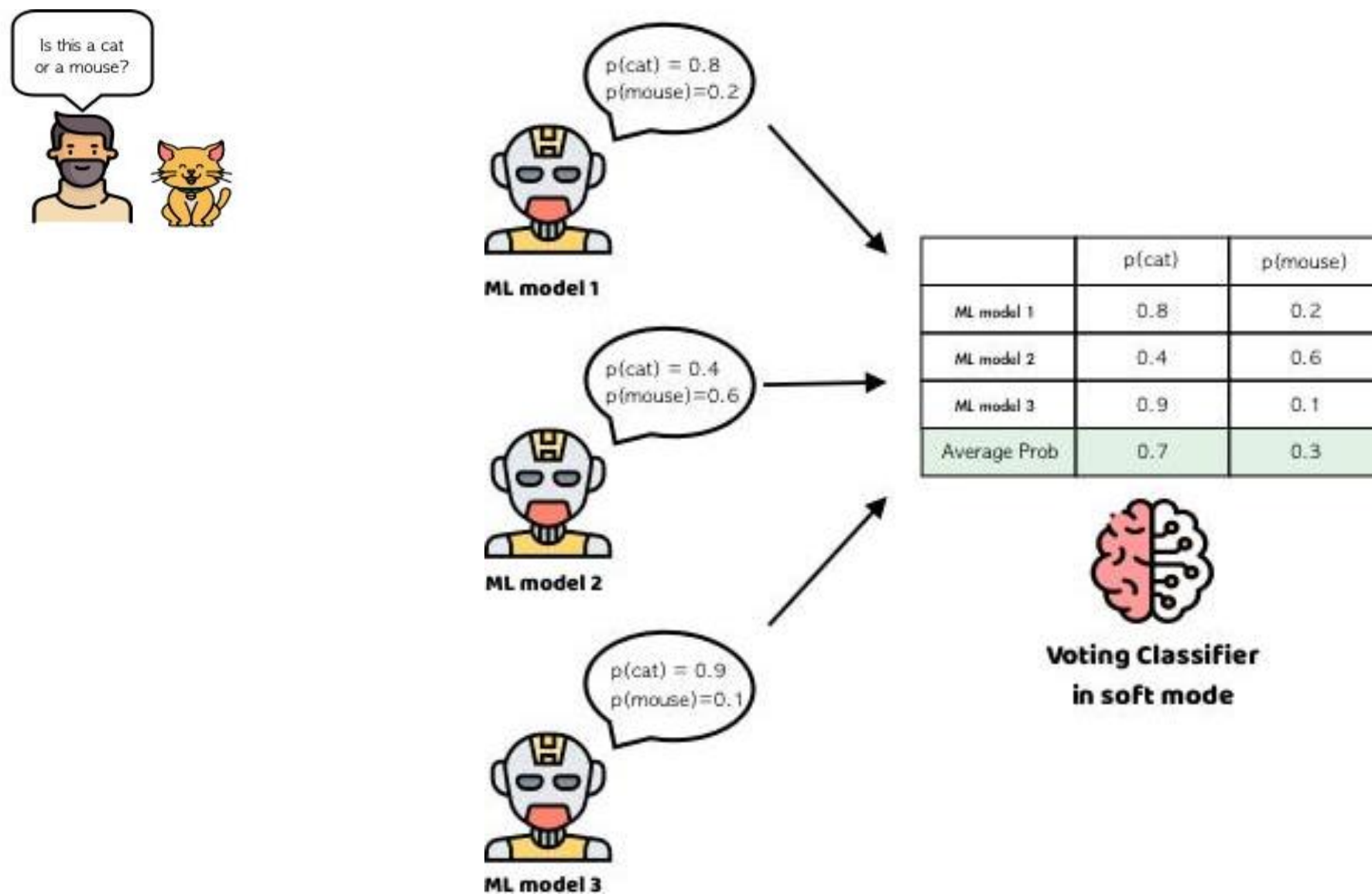
Soft Voting:

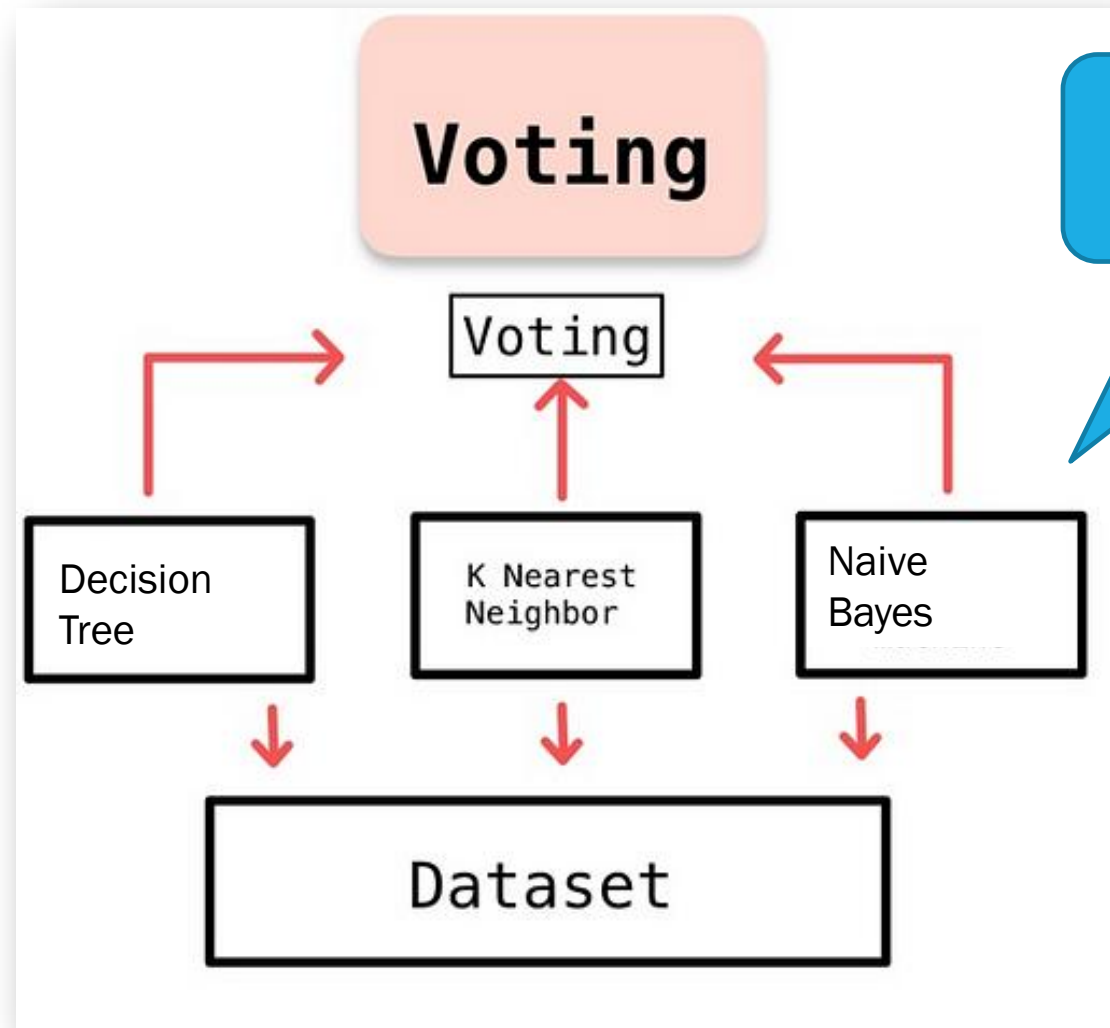
In regression tasks, the final prediction is typically obtained by averaging the predictions of the base models.

Or classification with probabilistic approaches

Also called weighted average ensemble model







We usually want diverse classifiers

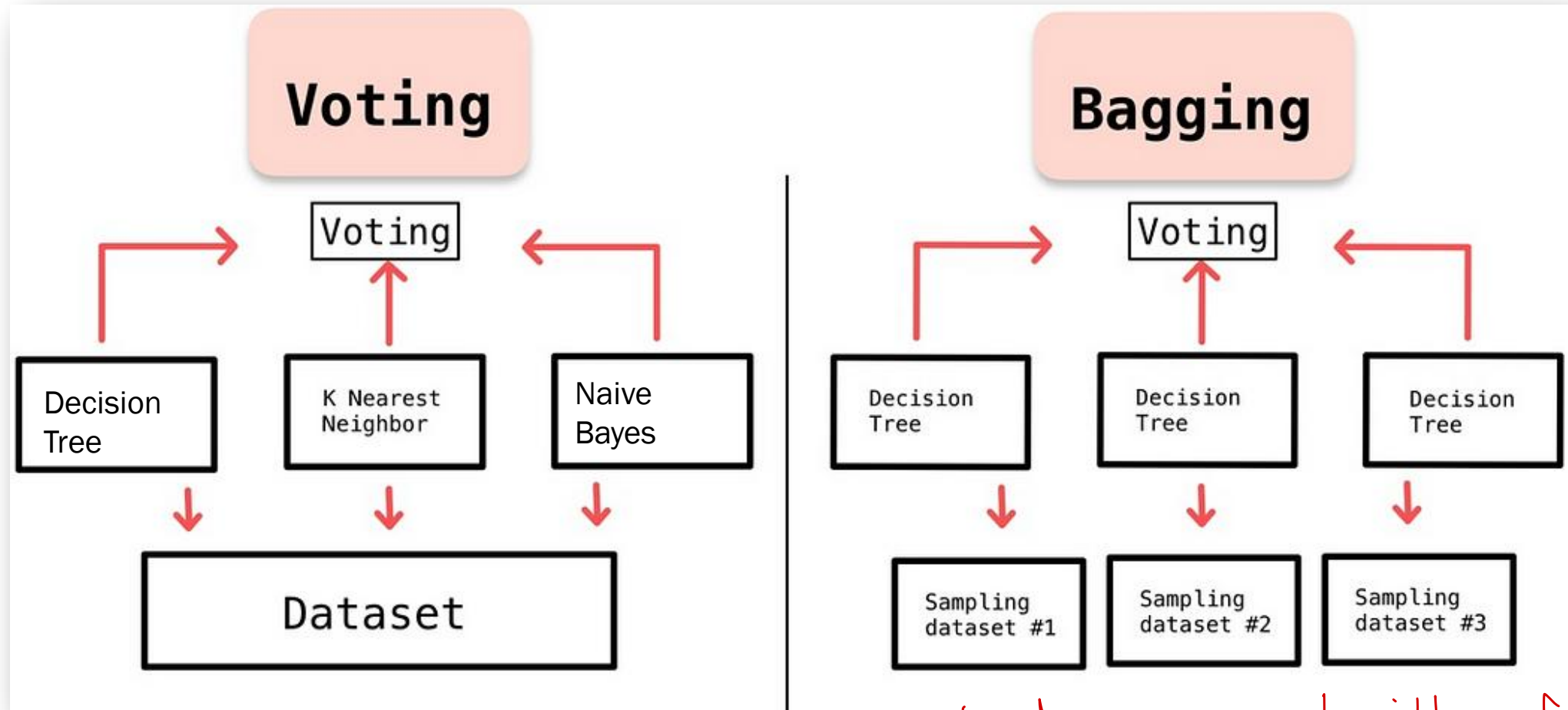
1. Voting
2. Bagging
3. Stacking and Blending
4. Boosting

Bagging

Definition

Bagging (Bootstrap Aggregating)

Bagging is an ensemble technique that aims to reduce variance and prevent overfitting by training multiple models independently and then averaging their predictions. Bagging works by generating different training datasets through bootstrapping and training a separate model on each of these datasets.

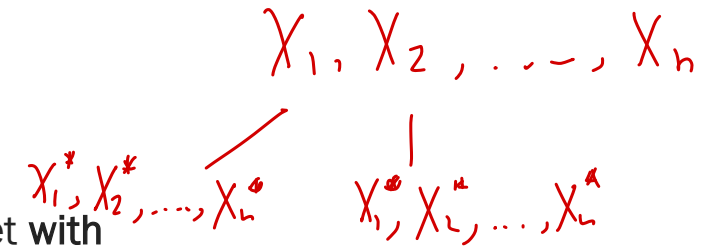


vote based on same algorithm from bootstrapped datasets

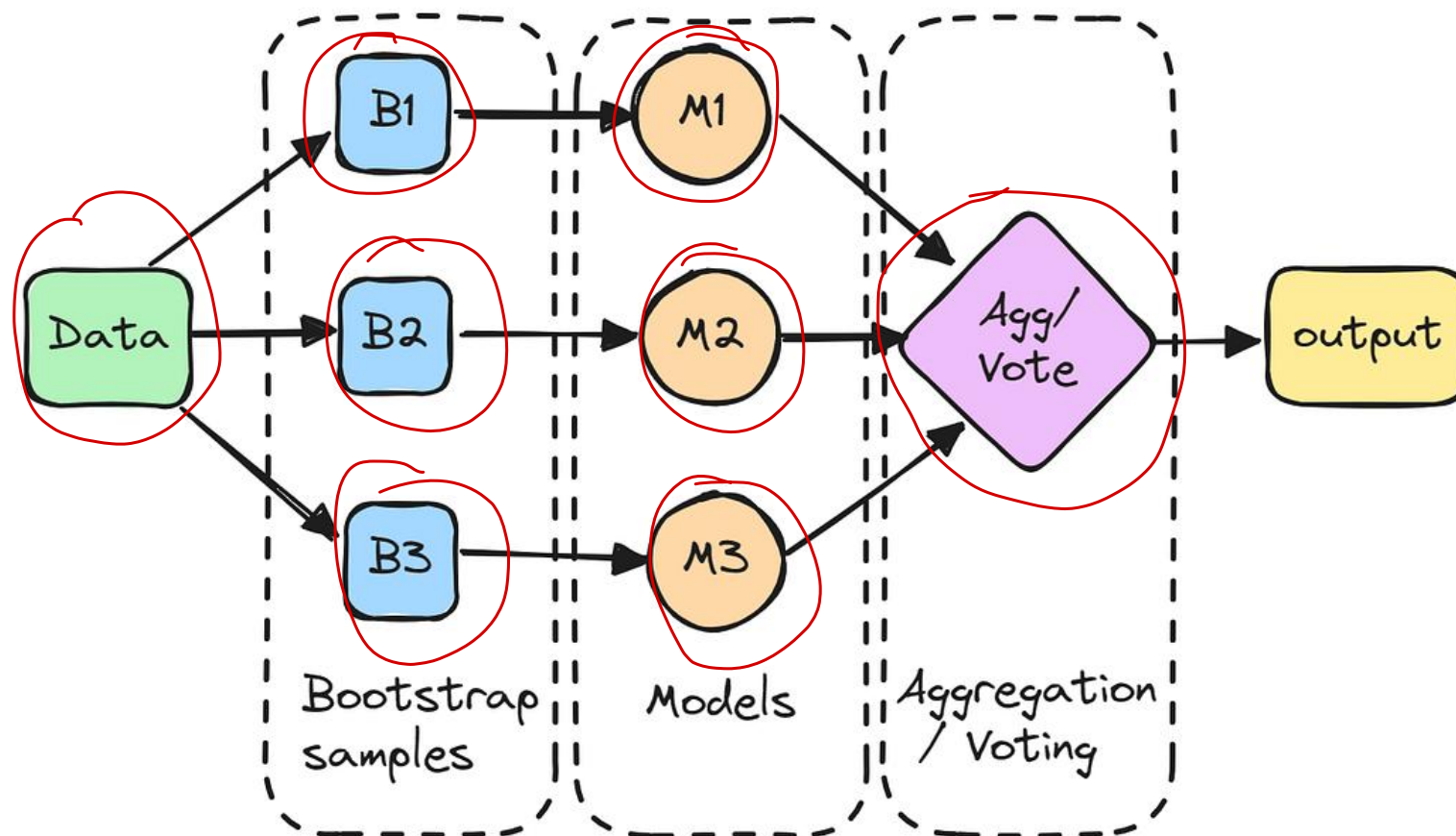
Algorithm

How Bagging Works:

- 1. Bootstrap Sampling:** Randomly sample data from the training set **with replacement** to create multiple subsets (bootstrap samples). Some data points may be repeated, while others might be omitted in each sample.
- 2. Train Models:** A separate model (usually a weak learner like a decision tree) is trained on each bootstrap sample.
- 3. Aggregate Predictions:** For classification tasks, the predictions from all the models are combined using **majority voting**. For regression tasks, the models' predictions are averaged.

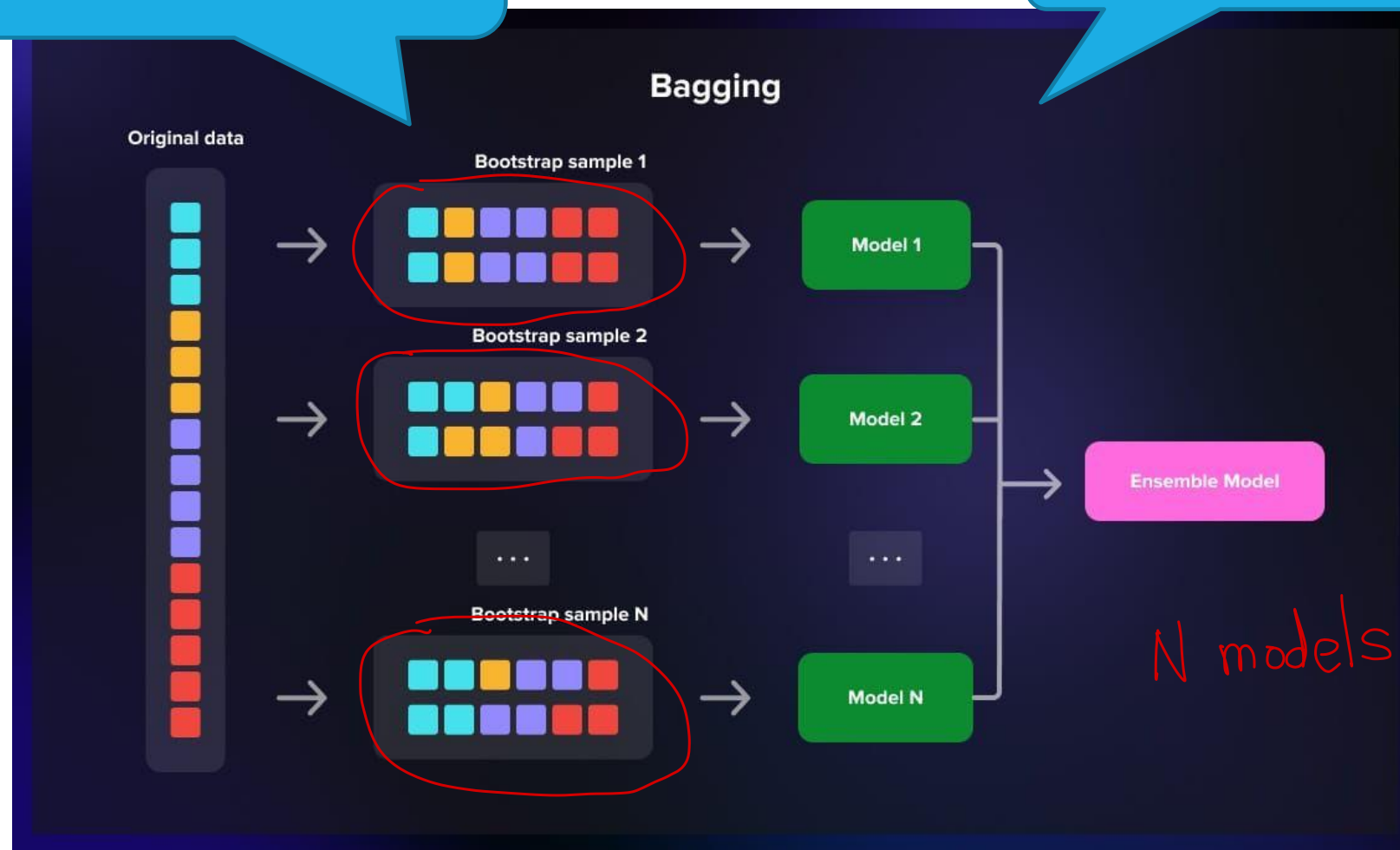


1. Bootstrap Sampling
2. Train Models
3. Aggregate Predictions



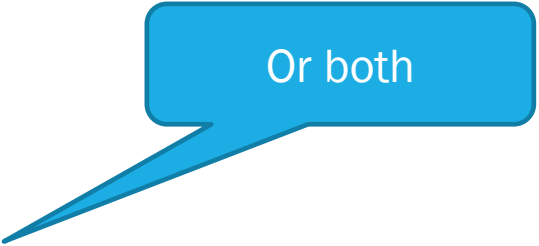
Some datapoints won't be selected, others will be present many times. All samples are of the same size.

All models are of same type



Sampling Rows: Standard Bootstrapping

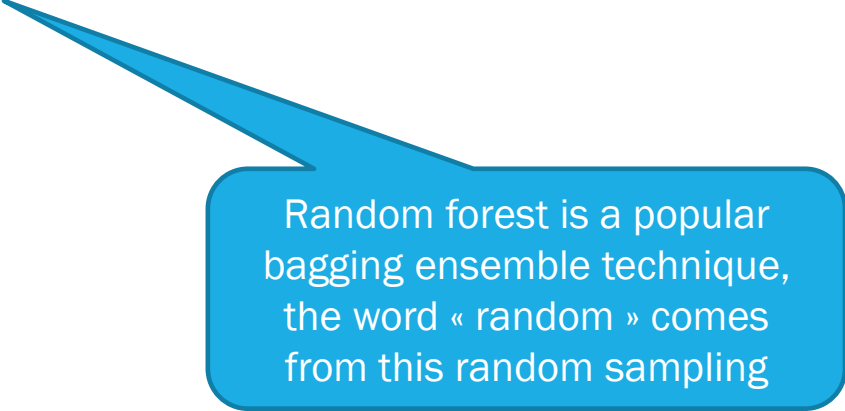
- In classical bootstrapping (like in Bagging), we randomly sample **rows** with replacement to create subsets of the dataset.
- Each subset may have repeated rows and will typically be of the same size as the original dataset (or smaller).



Or both

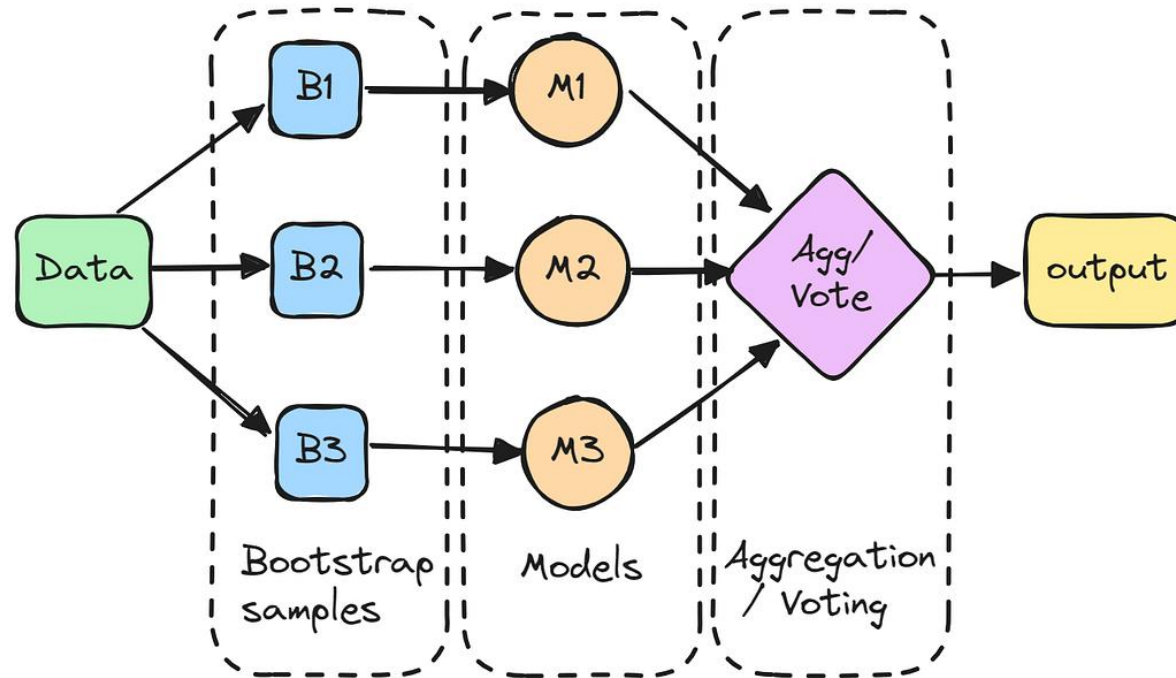
Sampling Columns: Feature Subsetting

- In some methods (like Random Forest), we can also sample a **subset of features (columns)** to use in the model.
- This is often done to introduce additional randomness and reduce the chance of overfitting.



Random forest is a popular bagging ensemble technique, the word « random » comes from this random sampling

Train Models: Train a separate model (e.g., a decision tree) on each of the bootstrapped datasets.



Aggregate Predictions: After all models are trained, combine their predictions (e.g., by majority voting in classification tasks or averaging in regression tasks) to make the final prediction.

Example

Everything about Random Forest



Abhishek Jain · Follow
4 min read · Sep 14, 2024

[Source](#)

ENSEMBLE LEARNING

Random Forest, Explained: A Visual Guide with Code Examples

Making tree-mendous predictions with random trees

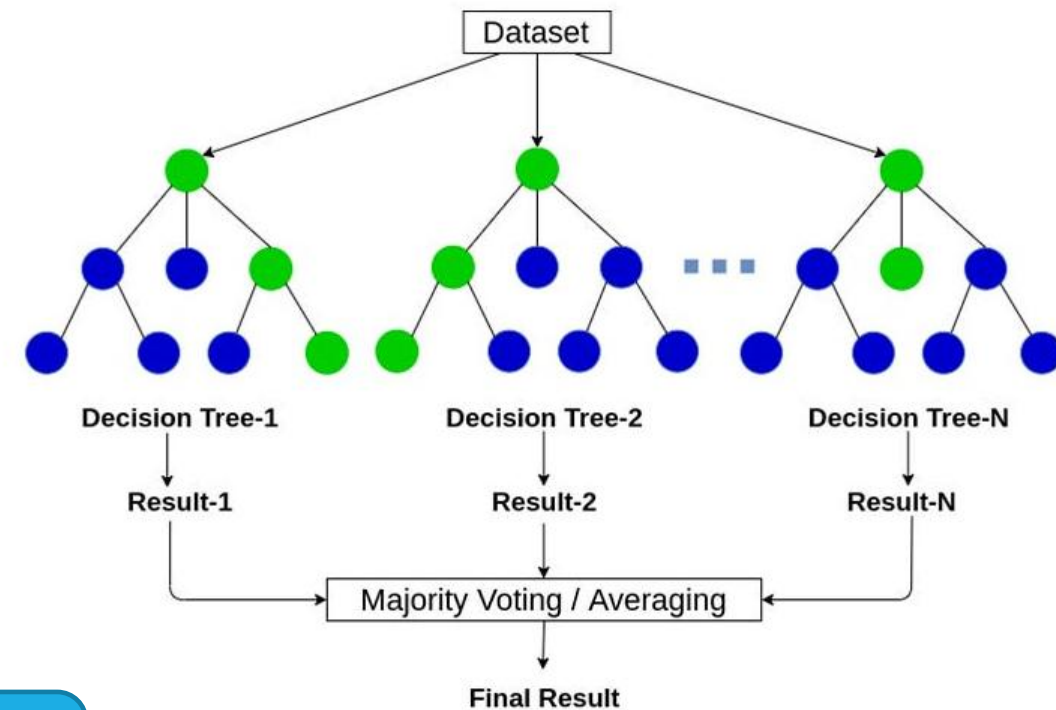


Samy Baladram · Follow
Published in TDS Archive · 12 min read · Nov 7, 2024

[Source](#)

The PlayGolf example we had for decision trees

Random Forest



Numerical
Temperature
Humidity

Categorical

Sunny : 0/1
Overcast: 0/1
Rainy: 0/1

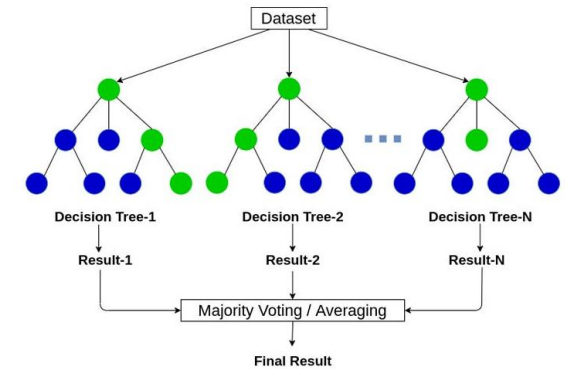
Windy: 0/1

One-hot encoding
(mutually exclusive)

Training Set

0	1	0	0	85	85	0	No
1	1	0	0	80	90	1	No
2	0	1	0	83	78	0	Yes
3	0	0	1	70	96	0	Yes
4	0	0	1	68	80	0	Yes
5	0	0	1	65	70	1	No
6	0	1	0	64	65	1	Yes
7	1	0	0	72	95	0	No
8	1	0	0	69	70	0	Yes
9	0	0	1	75	10	0	Yes
10	1	0	0	75	70	1	Yes
11	0	1	0	72	90	1	Yes
12	0	1	0	81	75	0	Yes
13	0	0	1	71	80	1	No

Random Forest



Random Forest Classifier

```
class sklearn.ensemble.RandomForestClassifier(n_estimators=100,  
*, criterion='gini', max_depth=None, min_samples_split=2,  
min_samples_leaf=1, min_weight_fraction_leaf=0.0,  
max_features='sqrt', max_leaf_nodes=None, min_impurity_decrease=0.0,  
bootstrap=True, oob_score=False, n_jobs=None, random_state=None,  
verbose=0, warm_start=False, class_weight=None, ccp_alpha=0.0,  
max_samples=None, monotonic_cst=None)
```

max_features (also part of
DecisionTreeClassifier, but important here!)

Many parameters are the same
as for Decision Tree Classifier

Extra parameter for
bootstrap (T/F)

n_estimators, the number of trees

max_samples (if bootstrap true)

Source

Key Differences Between Bagging and Voting:

Aspect	Bagging	Voting
Dataset Subsets	Models are trained on different subsets of data (via bootstrapping)	Models are trained on the same dataset
Model Diversity	Typically uses the same type of model (e.g., decision trees)	Often uses different types of models (e.g., tree, SVM, logistic regression)
Training Process	Uses bootstrapping to generate diverse training data	No bootstrapping, models use the same full dataset
Aggregation Method	Uses majority voting (classification) or averaging (regression) for final prediction	Uses majority voting (classification) or averaging (regression)
Purpose	Primarily aims to reduce variance and overfitting by averaging models trained on different datasets	Aims to leverage the strengths of different models to improve generalization
Examples	Random Forest (multiple decision trees)	Combining SVM, decision tree, and logistic regression

Good comparison (from GenAI)

1. Voting
2. Bagging
3. Stacking and Blending
4. Boosting

Stacking and Blending

Definition

Stacking is a sophisticated ensemble technique that involves combining different types of models (often called base learners) to improve performance.

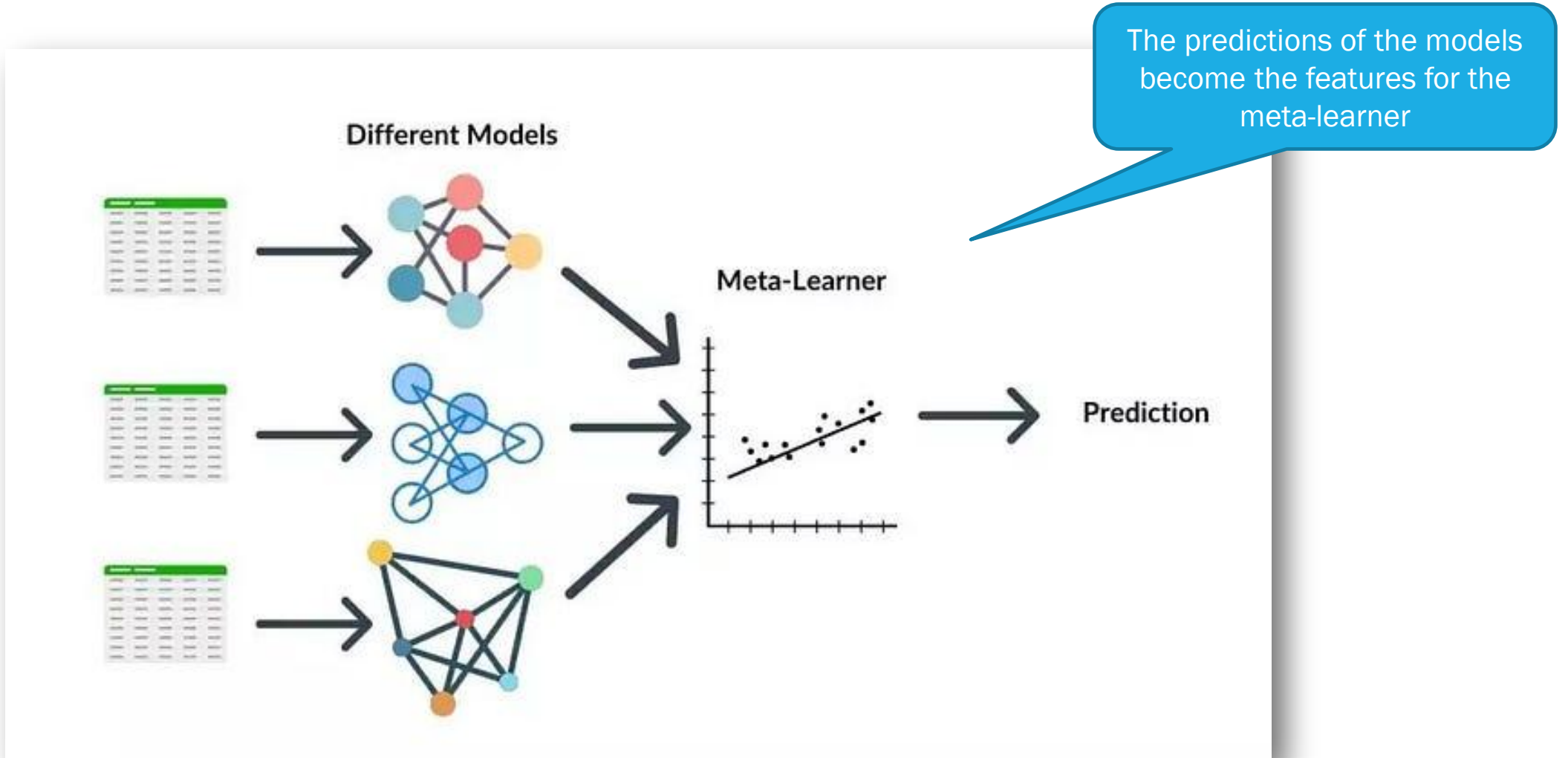
The idea behind stacking is to leverage the strengths of several models by training a **meta-model** (often called a second-level model) that learns to make predictions based on the outputs of the base models.



Algorithm

How Stacking Works:

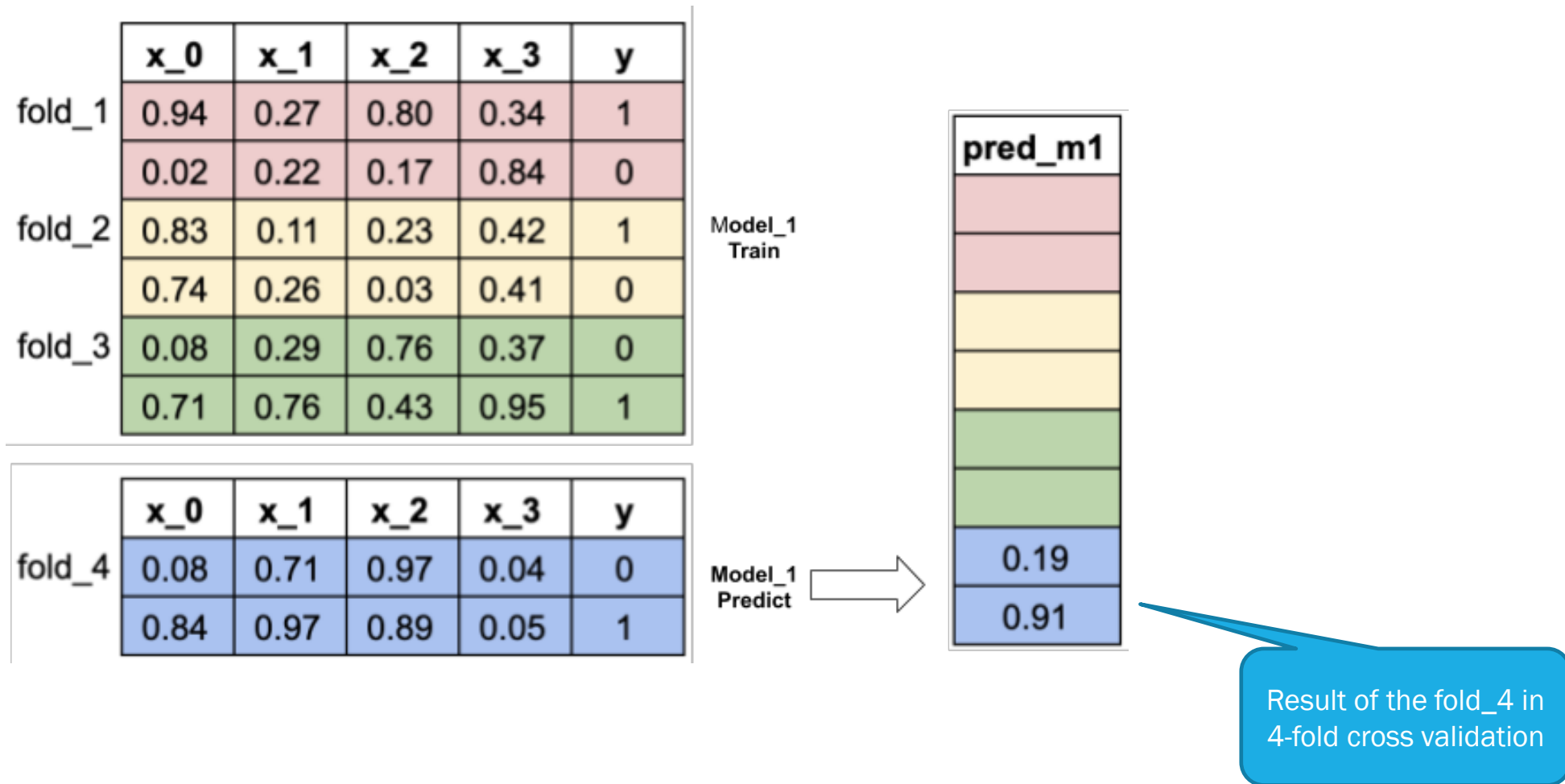
1. Train multiple base models (e.g., decision trees, logistic regression, SVMs) on the training data.
2. The predictions from these base models are fed into a **meta-model** (either a classification model like a neural network or a regression model).
3. The meta-model learns to combine the predictions of the base models and outputs the final prediction.



Walk through

train_data						
		x_0	x_1	x_2	x_3	y
fold_1		0.94	0.27	0.80	0.34	1
		0.02	0.22	0.17	0.84	0
fold_2		0.83	0.11	0.23	0.42	1
		0.74	0.26	0.03	0.41	0
fold_3		0.08	0.29	0.76	0.37	0
		0.71	0.76	0.43	0.95	1
fold_4		0.08	0.71	0.97	0.04	0
		0.84	0.97	0.89	0.05	1

test_data					
x_0	x_1	x_2	x_3	y	
0.74	0.17	0.820	0.31	1	
0.04	0.27	0.13	0.80	0	
0.87	0.10	0.24	0.39	1	



train_data

	x_0	x_1	x_2	x_3	y
fold_1	0.94	0.27	0.80	0.34	1
	0.02	0.22	0.17	0.84	0
fold_2	0.83	0.11	0.23	0.42	1
	0.74	0.26	0.03	0.41	0
fold_3	0.08	0.29	0.76	0.37	0
	0.71	0.76	0.43	0.95	1
fold_4	0.08	0.71	0.97	0.04	0
	0.84	0.97	0.89	0.05	1

new_features

pred_m1
0.96
0.03
0.90
0.12
0.03

train_data

	x_0	x_1	x_2	x_3	y
fold_1	0.94	0.27	0.80	0.34	1
	0.02	0.22	0.17	0.84	0
fold_2	0.83	0.11	0.23	0.42	1
	0.74	0.26	0.03	0.41	0
fold_3	0.08	0.29	0.76	0.37	0
	0.71	0.76	0.43	0.95	1
fold_4	0.08	0.71	0.97	0.04	0
	0.84	0.97	0.89	0.05	1

pred_m2

0.86
0.13
0.90
0.11
0.08
0.67
0.20
0.90

train_data

	x_0	x_1	x_2	x_3	y
fold_1	0.94	0.27	0.80	0.34	1
	0.02	0.22	0.17	0.84	0
fold_2	0.83	0.11	0.23	0.42	1
	0.74	0.26	0.03	0.41	0
fold_3	0.08	0.29	0.76	0.37	0
	0.71	0.76	0.43	0.95	1
fold_4	0.08	0.71	0.97	0.04	0
	0.84	0.97	0.89	0.05	1

pred_m3

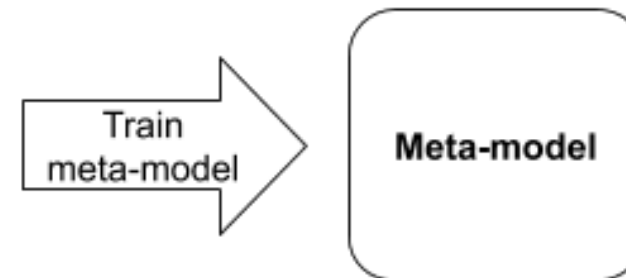
0.66
0.30
0.85
0.11
0.08
0.67
0.20
0.90

Repeat the 4-fold cross-validation on 2 other models

The outputs of these models are then used as features for a meta-model (e.g., a logistic regression), which makes the final classification decision.

New train_data for meta-model

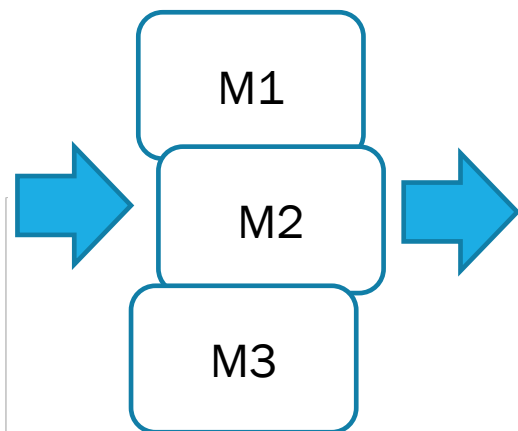
pred_m1	pred_m2	pred_m3	y
0.96	0.86	0.66	1
0.03	0.13	0.30	0
0.90	0.90	0.85	1
0.12	0.10	0.11	0
0.03	0.09	0.08	0
0.77	0.50	0.67	1
0.19	0.65	0.20	0
0.91	0.89	0.90	1



The predictions of the models become the features for the meta-learner

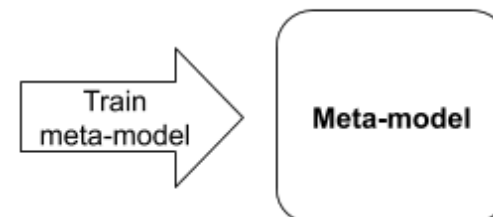
train_data

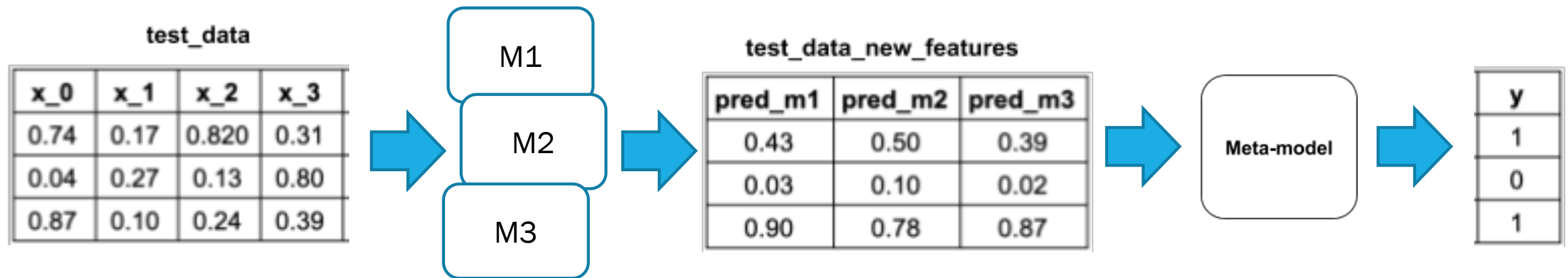
x_0	x_1	x_2	x_3	y
0.94	0.27	0.80	0.34	1
0.02	0.22	0.17	0.84	0
0.83	0.11	0.23	0.42	1
0.74	0.26	0.03	0.41	0
0.08	0.29	0.76	0.37	0
0.71	0.76	0.43	0.95	1
0.08	0.71	0.97	0.04	0
0.84	0.97	0.89	0.05	1



New train_data for meta-model

pred_m1	pred_m2	pred_m3	y
0.96	0.86	0.66	1
0.03	0.13	0.30	0
0.90	0.90	0.85	1
0.12	0.10	0.11	0
0.03	0.09	0.08	0
0.77	0.50	0.67	1
0.19	0.65	0.20	0
0.91	0.89	0.90	1



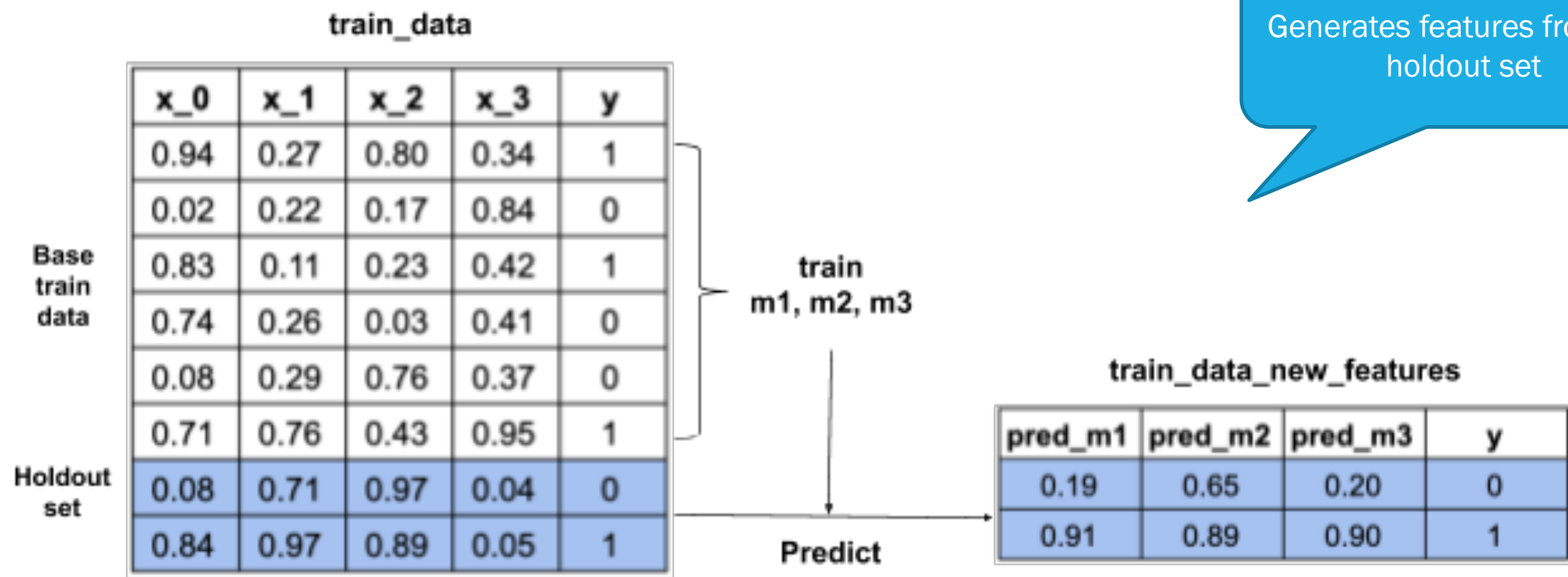


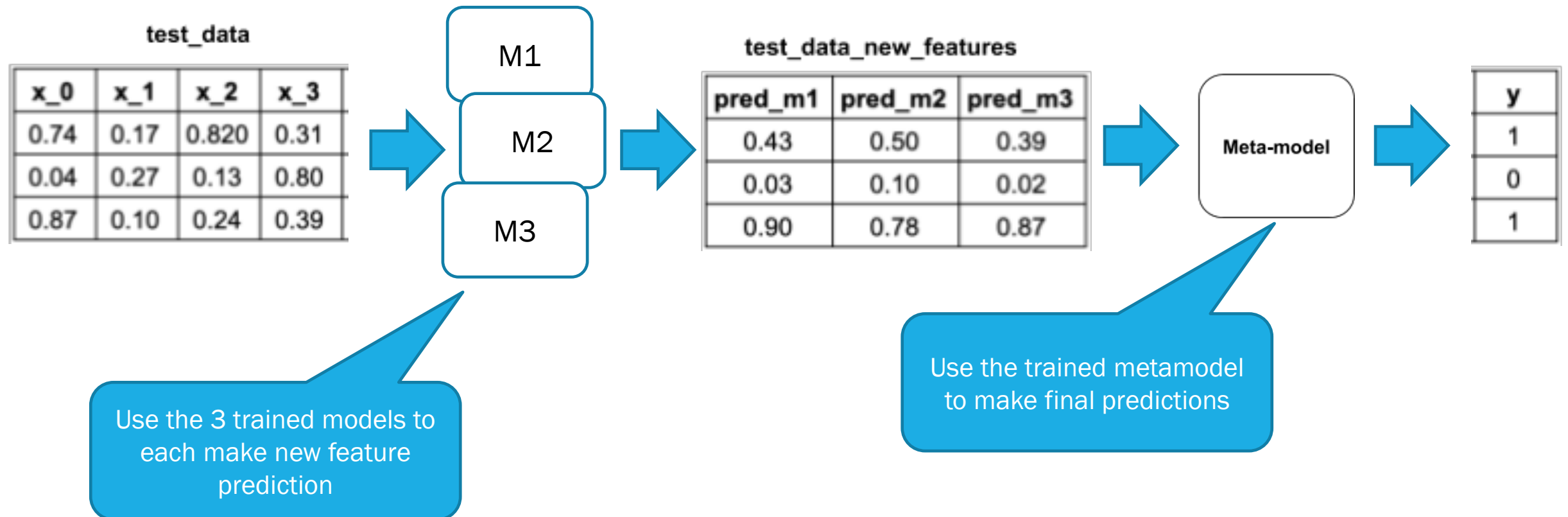
The title 'Blending' is centered within a dark gray rectangular box. This box is positioned on top of a larger, light blue rectangular area that has a stepped, L-shaped appearance on its right side.

Blending

Blending is very similar to Stacking. It also uses base models to provide base predictions as new features and a new meta model is trained on the new features that gives the final prediction.

The only difference is that training of the meta-model is applied on a separate holdout set (e.g 10% of train_data) rather on full and folded training set.





1. Voting
2. Bagging
3. Stacking and Blending
4. Boosting

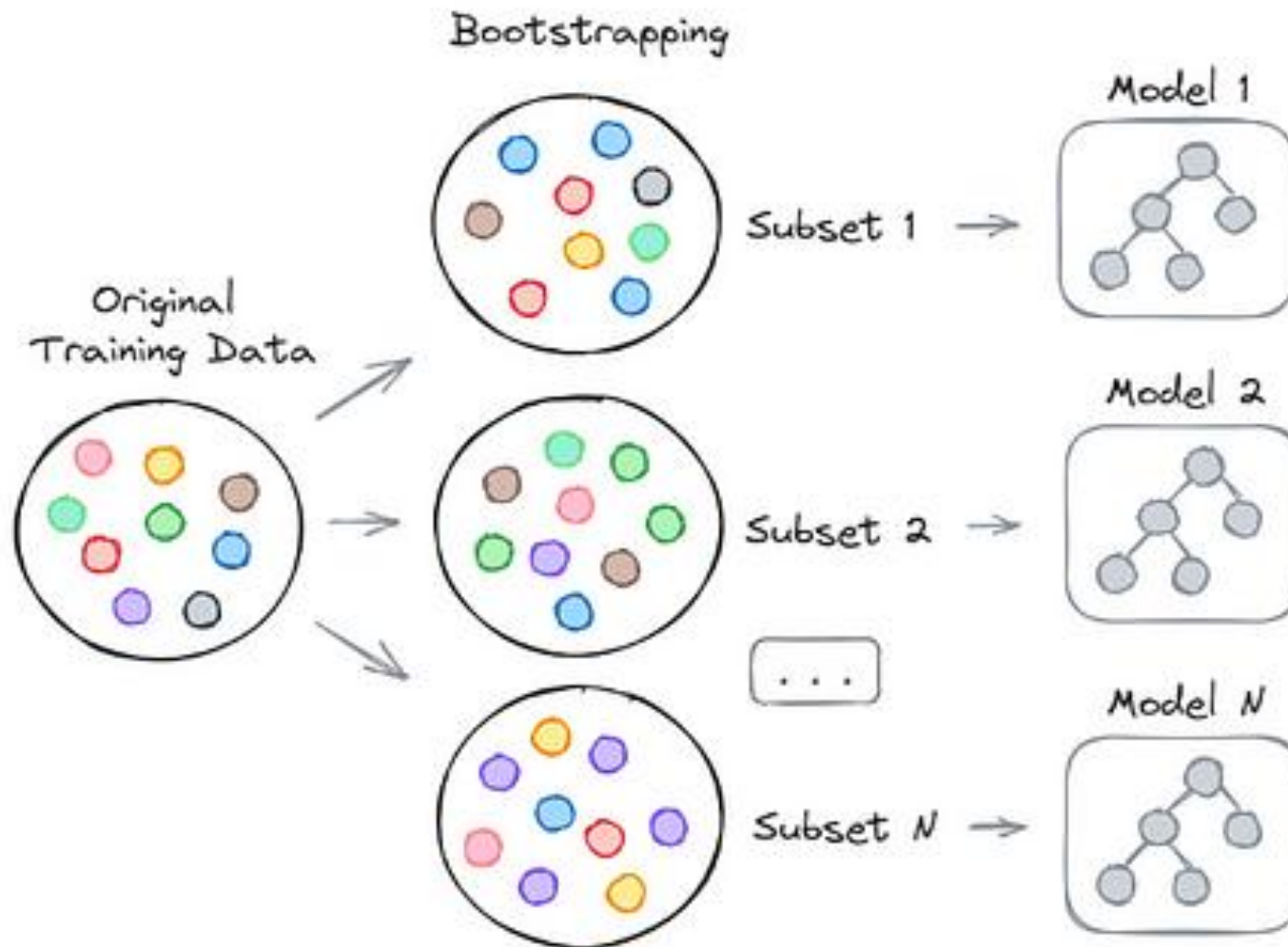
Boosting



Definition

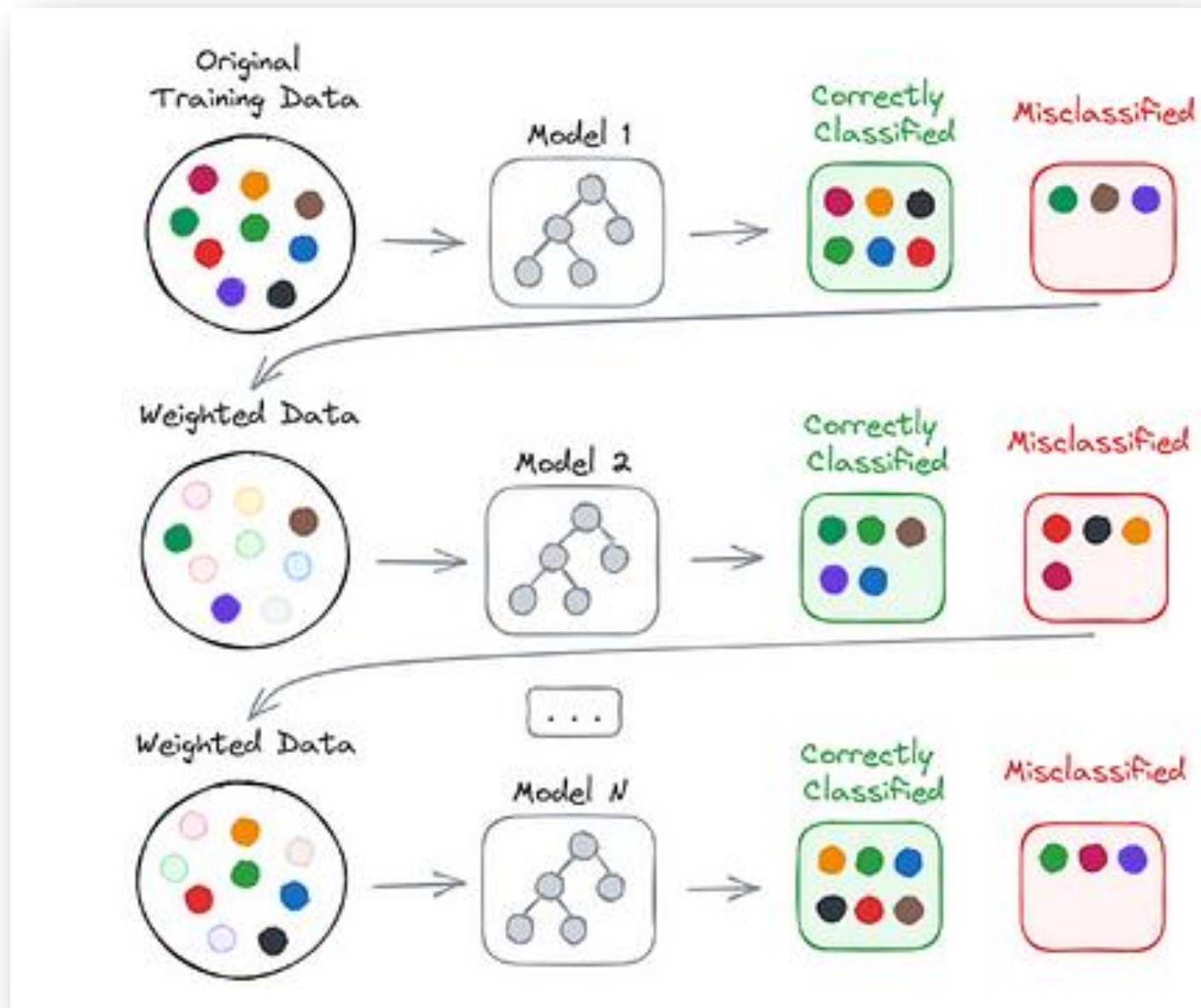
Boosting is another ensemble technique that focuses on **reducing both bias and variance** by training models sequentially, where each subsequent model attempts to correct the errors of the previous ones.

Unlike bagging, where models are trained independently, boosting builds models iteratively.



BAGGING

BOOSTING

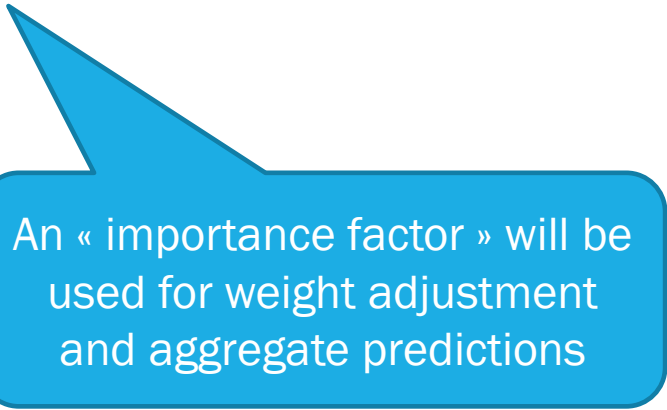




Algorithm

How Boosting Works:

- 1. Train Weak Learners Sequentially:** Models are trained one after another, and each model pays more attention to the data points that were misclassified by previous models.
- 2. Adjust Weights:** In each iteration, the weights of incorrectly predicted samples are increased, so subsequent models focus more on those difficult-to-predict samples.
- 3. Aggregate Predictions:** The final prediction is made by taking a weighted average or sum of the individual model predictions.



An « importance factor » will be used for weight adjustment and aggregate predictions

AdaBoost Illustrated



Dr. Roi Yehoshua · [Follow](#)

Published in AI Made Simple · 12 min read · Mar 8, 2023

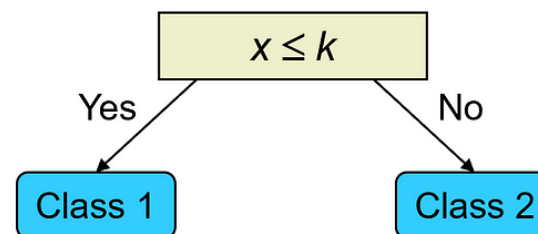
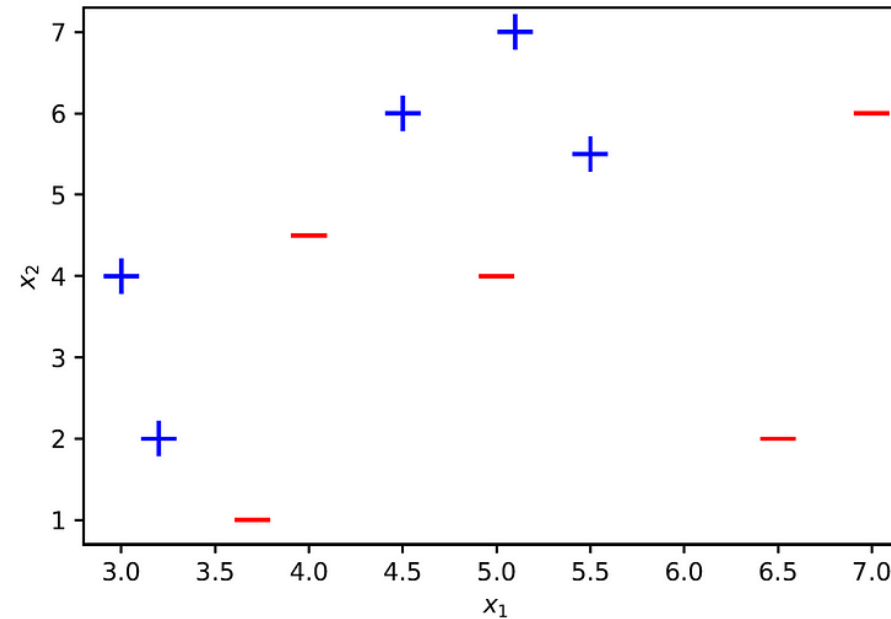
[Source](#)

AdaBoost (short for Adaptive Boosting) is a powerful boosting algorithm that can boost the performance of any machine learning model. It was introduced in 1995 by Freund and Shapira [1], who won the 2003 Gödel Prize (one of the most prestigious prizes in Computer Science) for their work. It is still one of the most popular boosting algorithms used today.

If you hear « Gradient boosting »... that's a different algorithm. Here, I'll only focus on Adaptive Boosting

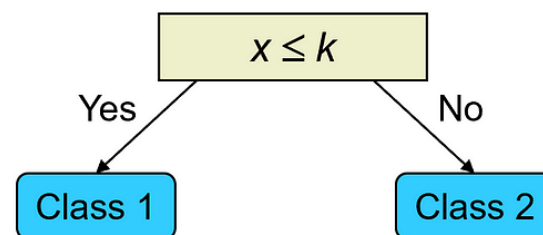
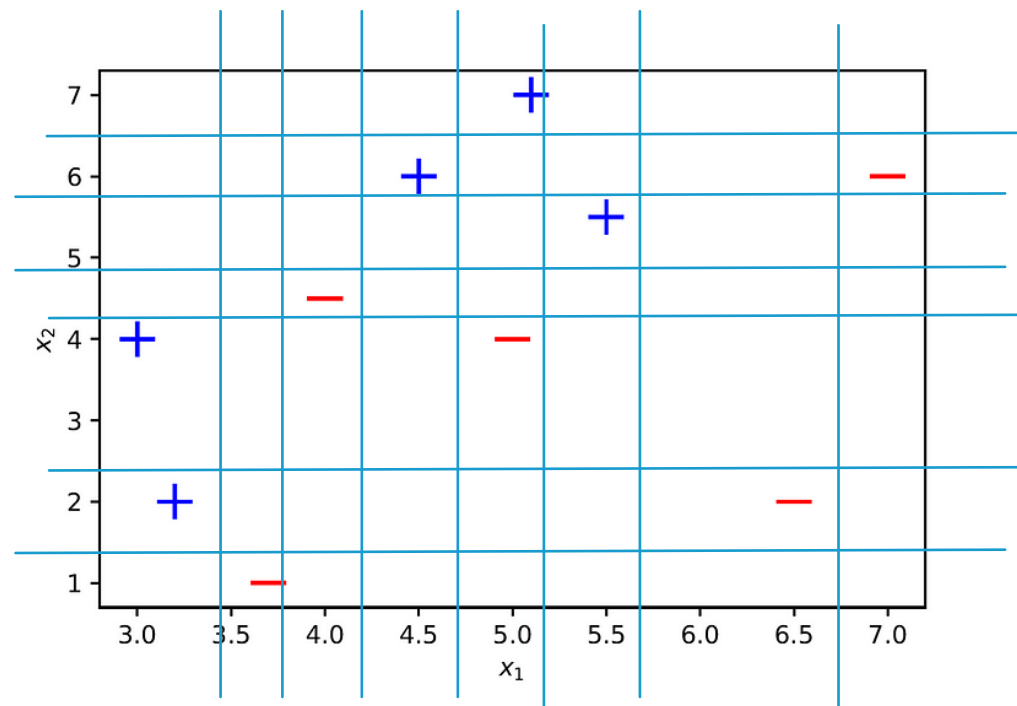
Illustration with decision trees.

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1
6	5	4	-1
7	5.1	7	1
8	5.5	5.5	1
9	6.5	2	-1
10	7	6	-1



Weak learner is a decision « stump »

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1
6	5	4	-1
7	5.1	7	1
8	5.5	5.5	1
9	6.5	2	-1
10	7	6	-1



$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

sample	x1	x2	class	weight
1	3	4	1	0.1
2	3.2	2	1	0.1
3	3.7	1	-1	0.1
4	4	4.5	-1	0.1
5	4.5	6	1	0.1
6	5	4	-1	0.1
7	5.1	7	1	0.1
8	5.5	5.5	1	0.1
9	6.5	2	-1	0.1
10	7	6	-1	0.1

Gini can use the weights

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

Gini Index over the dataset

« -1 » = 5

« 1 » = 5

$$1 - (5/10)^2 - (5/10)^2 = 0,5$$

No weights

Gini Index over the dataset

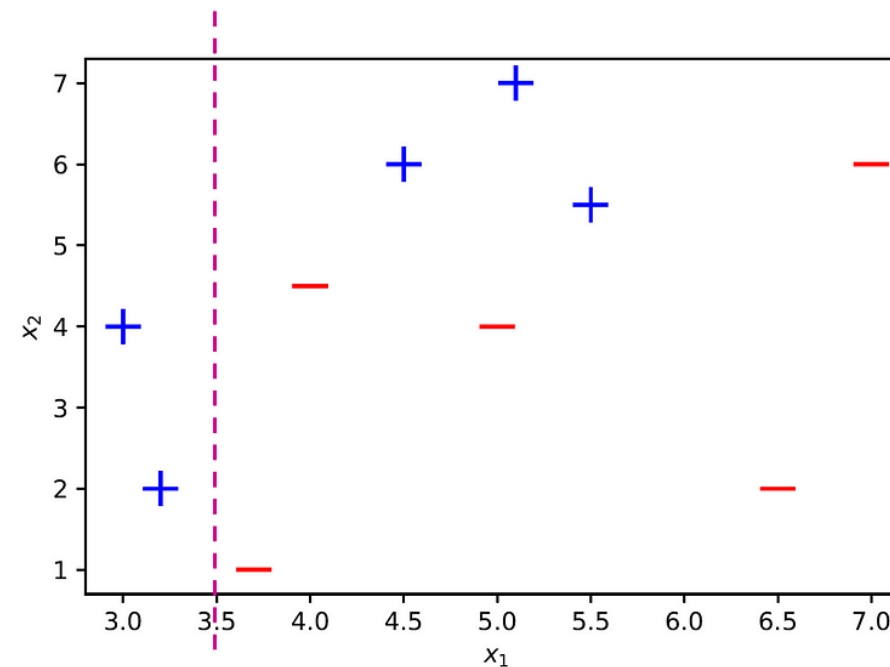
« -1 » = 0.1 + 0.1 + 0.1 + 0.1 + 0.1

« 1 » = 0.1 + 0.1 + 0.1 + 0.1 + 0.1

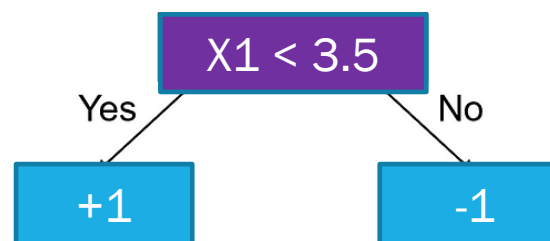
$$1 - (0.5/1)^2 - (0.5/1)^2 = 0,5$$

First boosting round

sample	x1	x2	class	weight
1	3	4	1	0.1
2	3.2	2	1	0.1
3	3.7	1	-1	0.1
4	4	4.5	-1	0.1
5	4.5	6	1	0.1
6	5	4	-1	0.1
7	5.1	7	1	0.1
8	5.5	5.5	1	0.1
9	6.5	2	-1	0.1
10	7	6	-1	0.1



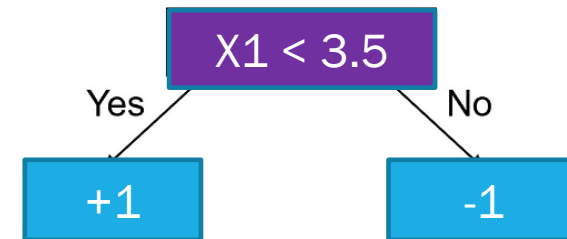
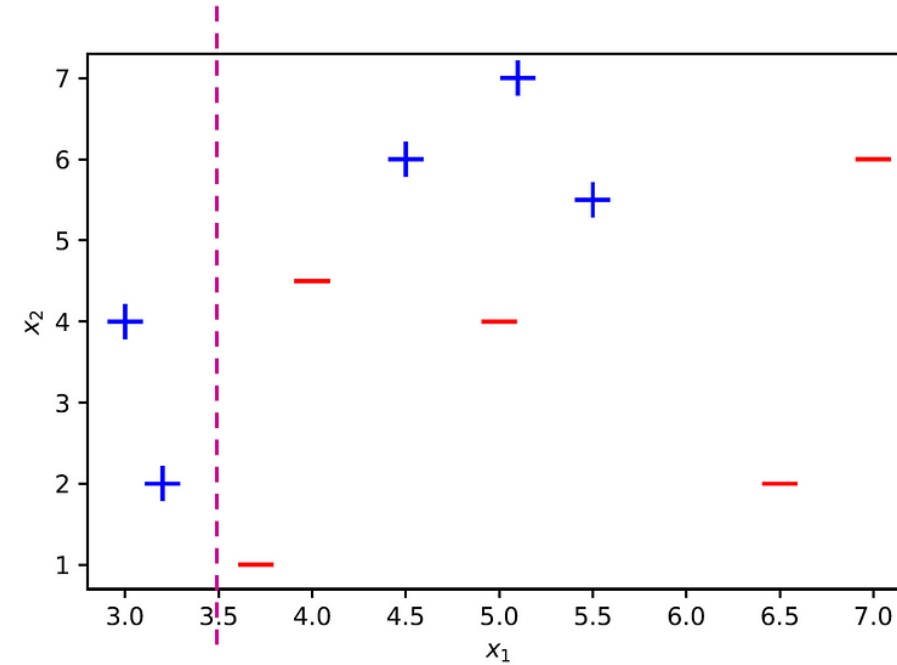
Calculate Gini Index at each split point on x_1 and x_2



Find the first stump

sample	x1	x2	class	weight
1	3	4	1	0.1
2	3.2	2	1	0.1
3	3.7	1	-1	0.1
4	4	4.5	-1	0.1
5	4.5	6	1	0.1
6	5	4	-1	0.1
7	5.1	7	1	0.1
8	5.5	5.5	1	0.1
9	6.5	2	-1	0.1
10	7	6	-1	0.1

Identify misclassified points



Calculate an importance factor

sample	x1	x2	class	weight
1	3	4	1	0.1
2	3.2	2	1	0.1
3	3.7	1	-1	0.1
4	4	4.5	-1	0.1
5	4.5	6	1	0.1
6	5	4	-1	0.1
7	5.1	7	1	0.1
8	5.5	5.5	1	0.1
9	6.5	2	-1	0.1
10	7	6	-1	0.1

Learning rate

$$\alpha_t = \eta \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

Importance Factor

hyperparameter

Sum of errors

$$\alpha_1 = 0.5 \ln \left(\frac{1 - 0.3}{0.3} \right) = 0.424$$

pick learning rate

weight sum of misclassified

Weighted sum of misclassified

Readjust the weights of each data point

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

Use the importance factor to calculate the new weights

sample	x1	x2	class	weight
1	3	4	1	0.1
2	3.2	2	1	0.1
3	3.7	1	-1	0.1
4	4	4.5	-1	0.1
5	4.5	6	1	0.1
6	5	4	-1	0.1
7	5.1	7	1	0.1
8	5.5	5.5	1	0.1
9	6.5	2	-1	0.1
10	7	6	-1	0.1

$$w_{correct} = 0.1 \cdot e^{-0.424} = 0.065$$

$$w_{incorrect} = 0.1 \cdot e^{0.424} = 0.153$$

$$Z_1 = 3 \cdot 0.153 + 7 \cdot 0.065 = 0.917$$

3 misclassified

normalization

7 correctly classified

Readjust the weights of each data point

$$Z_1 = 3 \cdot 0.153 + 7 \cdot 0.065 = 0.917$$

Increase and decrease weights, then normalize

$$w_{incorrect} = \frac{0.153}{0.917} = 0.167$$

$$w_{correct} = \frac{0.065}{0.917} = 0.071$$

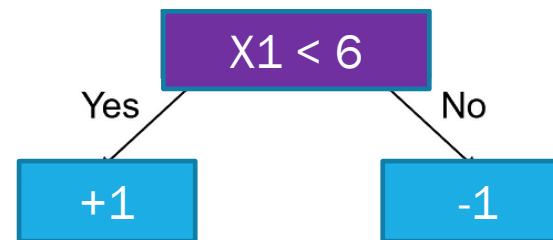
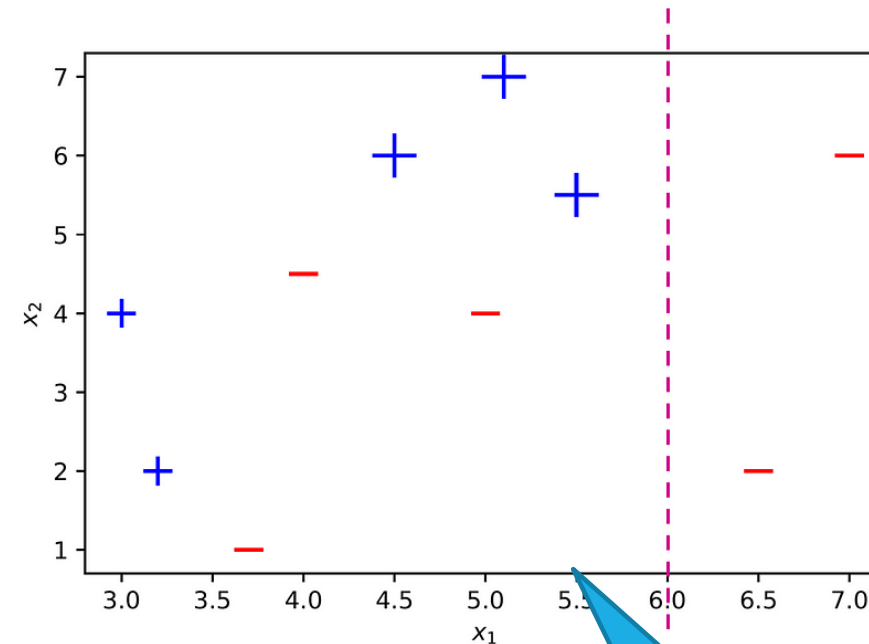
weights

sample	x1	x2	class	weight
1	3	4	1	0.071
2	3.2	2	1	0.071
3	3.7	1	-1	0.071
4	4	4.5	-1	0.071
5	4.5	6	1	0.167
6	5	4	-1	0.071
7	5.1	7	1	0.167
8	5.5	5.5	1	0.167
9	6.5	2	-1	0.071
10	7	6	-1	0.071

continue

Second boosting round

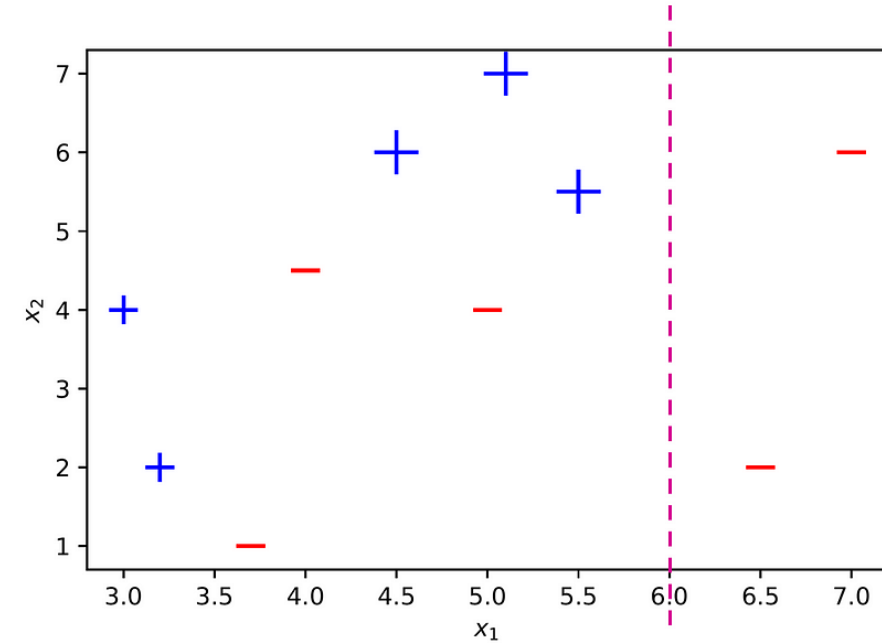
sample	x1	x2	class	weight
1	3	4	1	0.071
2	3.2	2	1	0.071
3	3.7	1	-1	0.071
4	4	4.5	-1	0.071
5	4.5	6	1	0.167
6	5	4	-1	0.071
7	5.1	7	1	0.167
8	5.5	5.5	1	0.167
9	6.5	2	-1	0.071
10	7	6	-1	0.071



Find the second stump

3 misclassified

sample	x1	x2	class	weight
1	3	4	1	0.071
2	3.2	2	1	0.071
3	3.7	1	-1	0.071
4	4	4.5	-1	0.071
5	4.5	6	1	0.167
6	5	4	-1	0.071
7	5.1	7	1	0.167
8	5.5	5.5	1	0.167
9	6.5	2	-1	0.071
10	7	6	-1	0.071



Identify misclassified points

Calculate importance factor

$$\alpha_2 = 0.5 \ln \left(\frac{1 - 0.213}{0.213} \right) = 0.65$$

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

$$\alpha_2 = 0.5 \ln \left(\frac{1 - 0.213}{0.213} \right) = 0.65$$

sample	x1	x2	class	weight
1	3	4	1	0.071
2	3.2	2	1	0.071
3	3.7	1	-1	0.071
4	4	4.5	-1	0.071
5	4.5	6	1	0.167
6	5	4	-1	0.071
7	5.1	7	1	0.167
8	5.5	5.5	1	0.167
9	6.5	2	-1	0.071
10	7	6	-1	0.071

$$W1 = 0.071 * e(-0.65) = 0.037$$

$$W2 = 0.071 * e(-0.65) = 0.037$$

$$W3 = 0.071 * e(0.65) = 0.136$$

$$W4 = 0.071 * e(0.65) = 0.136$$

$$W5 = 0.167 * e(-0.65) = 0.087$$

$$W6 = 0.071 * e(0.65) = 0.136$$

$$W7 = 0.167 * e(-0.65) = 0.087$$

$$W8 = 0.167 * e(-0.65) = 0.087$$

$$W9 = 0.071 * e(-0.65) = 0.037$$

$$W10 = 0.071 * e(-0.65) = 0.037$$

$$\alpha = 0.3$$

$$D(4) = 0.2$$

$$\frac{0.2 \exp(-0.3)}{4 \times 0.2 \exp(-0.3) + 0.2 \exp(0.3)} =$$

$$0.5 \log \left(\frac{1 - 0.4}{0.4} \right) = 0.5 \log \left(\frac{6}{4} \right) = 0.2027$$

$$Z = 3 \times 0.2 \exp(-0.4) + 2 \times 0.2 \exp(0.4)$$

$$\alpha = 0.4 \ln \left(\frac{1 - 0.2}{0.2} \right)$$

$$1 - 0.2^2 - 0.8^2 =$$

Sum (Z) is 0.8178

Readjust the weights of each data point

sample	x1	x2	class	weight
1	3	4	1	0.071
2	3.2	2	1	0.071
3	3.7	1	-1	0.071
4	4	4.5	-1	0.071
5	4.5	6	1	0.167
6	5	4	-1	0.071
7	5.1	7	1	0.167
8	5.5	5.5	1	0.167
9	6.5	2	-1	0.071
10	7	6	-1	0.071

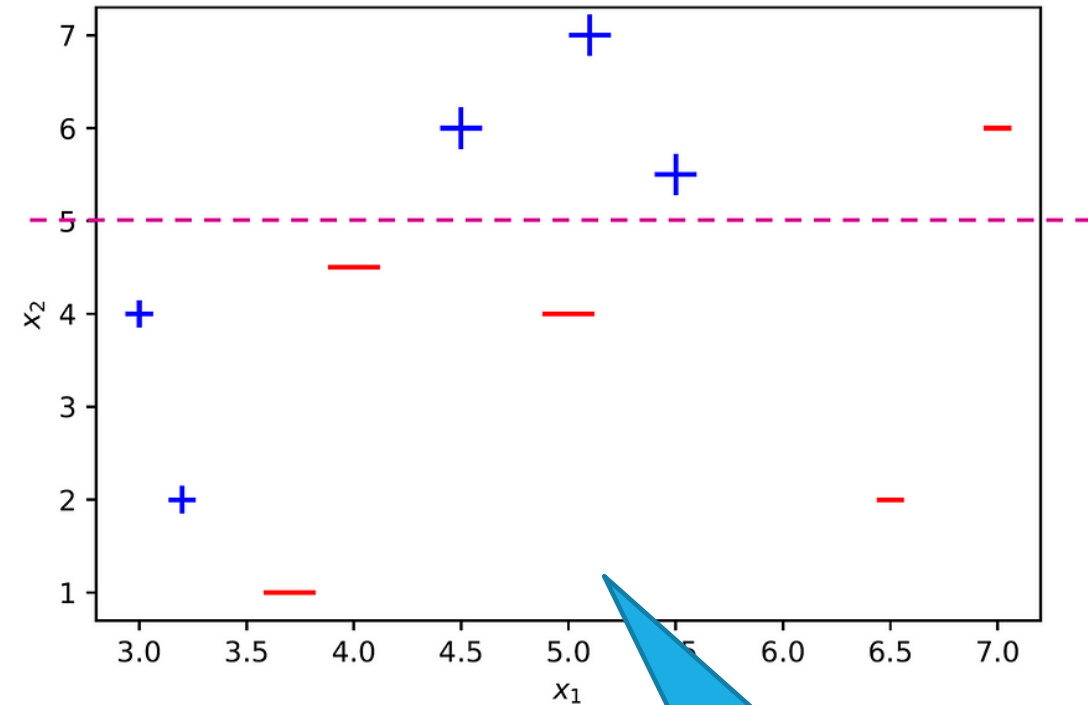


sample	x1	x2	class	weight
1	3	4	1	0.045
2	3.2	2	1	0.045
3	3.7	1	-1	0.167
4	4	4.5	-1	0.167
5	4.5	6	1	0.106
6	5	4	-1	0.167
7	5.1	7	1	0.106
8	5.5	5.5	1	0.106
9	6.5	2	-1	0.045
10	7	6	-1	0.045

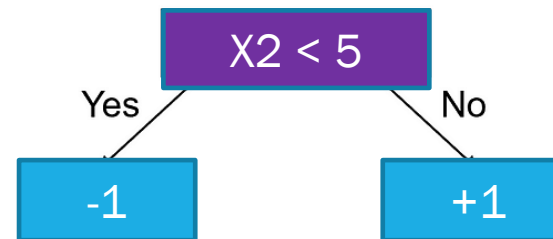
Use calculated
weights normalized

Third boosting round

	sample	x1	x2	class	weight	
	1	3	4	1	0.045	
	2	3.2	2	1	0.045	
	3	3.7	1	-1	0.167	
	4	4	4.5	-1	0.167	
	5	4.5	6	1	0.106	
	6	5	4	-1	0.167	
	7	5.1	7	1	0.106	
	8	5.5	5.5	1	0.106	
	9	6.5	2	-1	0.045	
	10	7	6	-1	0.045	



$$\alpha_3 = 0.5 \ln \left(\frac{1 - 0.135}{0.135} \right) = 0.923$$



Find the third stump

Round	1	2	3	4	5	6	7	8	9	10
1	1	1	-1	-1	-1	-1	-1	-1	-1	-1
2	1	1	1	1	1	1	1	1	-1	-1
3	-1	-1	-1	-1	1	-1	1	1	-1	1
Sum	0.15	0.15	-0.697	-0.697	1.149	-0.697	1.149	1.149	-1.996	-0.15
Sign	1	1	-1	-1	1	-1	1	1	-1	-1

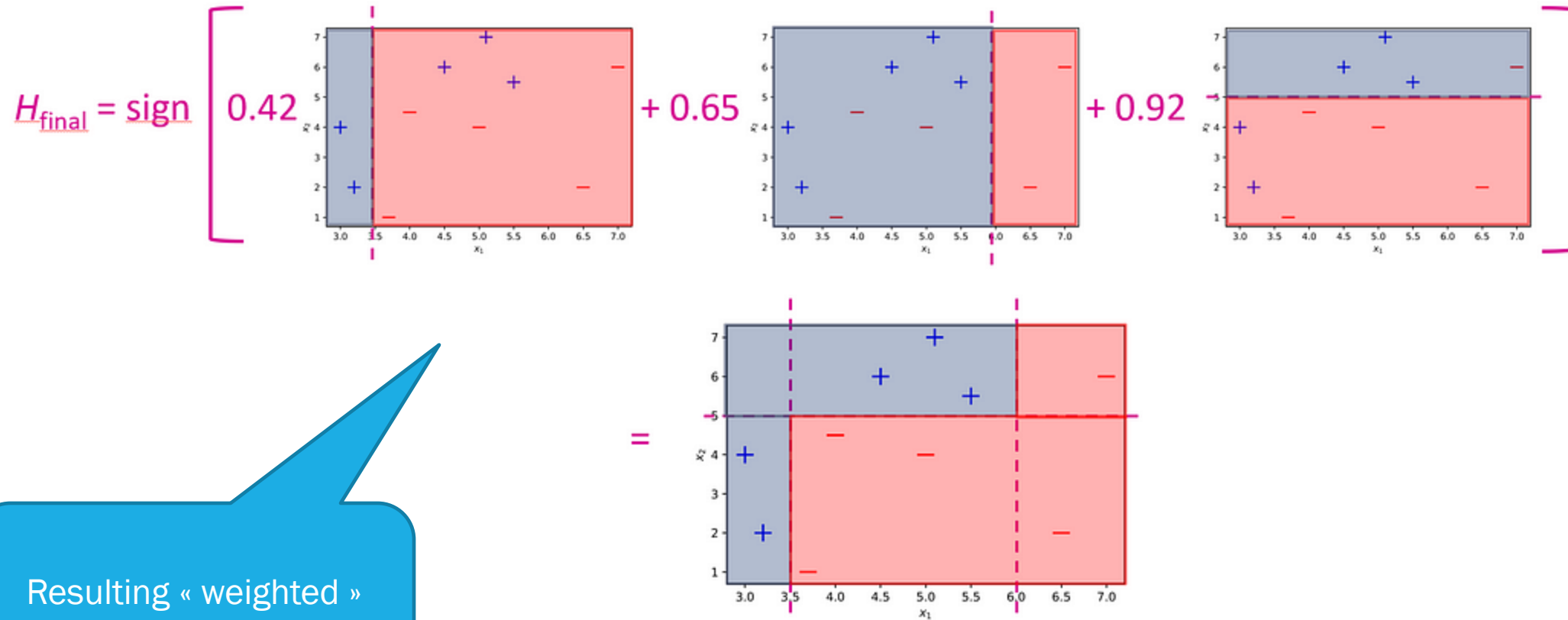
$$0.424 + 0.65 - 0.923 = 0.151$$

$$-0.424 + 0.65 - 0.923 = -0.697$$

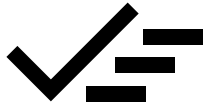
$$\alpha_1 = 0.424$$

$$\alpha_2 = 0.65$$

$$\alpha_3 = 0.923$$



Resulting « weighted »
decision boundaries



ENSEMBLE MODELS

- Introduction
- Ensemble approaches:
 - Voting
 - Bagging
 - Stacking and Blending
 - Boosting

